**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Debugging Syntax and Logic Errors

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. For valid ages 13-18, identify the two valid boundary values.

2. What is an IDE's editor used for?

A) Writing source code
B) Fixing physical CPU faults
C) Managing user access rights
D) Defragmenting a hard drive

3. "The first index in every OCR ERL array is 1." Correct this. Explain why.

4. A ticket program checks ages. "A few successful tests prove it has no faults." Correct this. How should it be tested?

## Today's objectives

Explain the core ideas of debugging syntax and logic errors using accurate OCR terminology. Apply a clear method to the worked and guided questions.

## Key vocabulary - Read, cover, recall

**Syntax error**
Code breaks language grammar and may fail to run.

**Logic error**
Code runs but behaves incorrectly.

**Trace and isolate**
Follow values and conditions to locate the first incorrect change.

**Repair and retest**
Correct the cause, then check both the original and adjacent boundary cases.

**Write one key term and its meaning in your own words:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Learn and practise

## New learning - Key ideas

| | |
|---|---|
| **Syntax error** | Code breaks language grammar and may fail to run. |
| **Logic error** | Code runs but behaves incorrectly. |
| **Trace and isolate** | Follow values and conditions to locate the first incorrect change. |
| **Repair and retest** | Correct the cause, then check both the original and adjacent boundary cases. |

## I Do - One worked example

**A loop uses total = total - value when calculating a sum. Diagnose and correct it.**

1. The program runs, so it may be a logic error.
2. Trace a small input to observe subtraction.
3. Replace subtraction with total = total + value.
4. Retest several inputs and compare totals with expected sums.

## We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A range check uses OR instead of AND for score >=0 and score <=100. Fix it.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Correct the Boolean operator that makes the range test accept every number.**

```
octet = 300
if octet >= 0 OR octet <= 255 then
print("Valid")
else
print("Invalid")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

## You Do - Independent exam questions

Q2. Differentiate a syntax error from a logic error. [2 marks]

Q3. Identify the fault in if age >= 13 OR age <= 18 for acceptance age 13-18. [2 marks]

Q4. Explain why testing after debugging is necessary. [2 marks]

## Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A logic error can occur in a program that runs without a syntax error. ☐ T ☐ F
2. Code breaks language grammar and may fail to run. ☐ T ☐ F
3. Code runs but behaves incorrectly. ☐ T ☐ F

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

## Do Now - Answers

**Q1** 13 and 18 lie on the edges of the valid range. ☐

**Q2** A - Writing source code ☐

**Q3** A fixed-size indexed collection of related elements; OCR ERL indexing begins at zero. ☐

**Q4** Test valid and invalid inputs. Compare actual and expected results to find faults. A few successful tests cannot prove there are no faults. ☐

## Code First - Model

```
octet = 300
if octet >= 0 AND octet <= 255 then
print("Valid")
else
print("Invalid")
endif
```

**Output: Invalid**

## We Do - Check the model

1. Retest -1, 0, 100 and 101. ☐

## Exit Ticket - Check

**1. TRUE 2. TRUE 3. TRUE**

## You Do - Self-mark

**Q2** Syntax violates language rules; logic produces incorrect behaviour even if program runs. ☐

**Q3** Use AND, not OR; OR incorrectly accepts values outside the range. ☐

**Q4** To verify the correction and ensure no new faults were introduced. ☐

## Try the question again

**1. Copy the FULL question you found most difficult:**

**2. In GREEN PEN, write a complete new answer:**